

Jaspreet Ranjit

4TH YEAR CS PHD CANDIDATE · UNIVERSITY OF SOUTHERN CALIFORNIA

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Research Statement

My research advances approaches for collaborating with, and evaluating AI systems in societally important, high-stakes domains, particularly in social services, public health, and mental health care, including examining the safety risks of deploying language models in these settings. I translate stakeholder requirements into open-source tools that convert unstructured text into structured insights for decision augmentation for community organizations in Los Angeles, and D.C. I also examine how AI agents will impact the future of work. My work has been recognized with the **Outstanding Paper Award at EMNLP 2024**, the **WiSE Merit Award for 2025**, the **Best Responsible AI Impact Award at SoCal AI Responsibility Summit**, and has received a **\$50K Min Family Challenge Grand Prize**.

Education

University of Southern California (USC)

Los Angeles, CA

PH.D. IN COMPUTER SCIENCE — NATURAL LANGUAGE PROCESSING

Present

Advisor: Swabha Swayamdipta, DILL Lab

Research Interests: NLP for social impact, Human-AI collaboration, Responsible AI

Leadership: Student leader of Center for AI in Society, Secretary of Women in Science and Engineering

University of Virginia (UVA)

Charlottesville, VA

MASTER OF SCIENCE IN COMPUTER SCIENCE

Advisor: Vicente Ordóñez

Thesis: *Analyzing Gender Biases in Visual Recognition Models*

University of Virginia (UVA)

Charlottesville, VA

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Advisor: Madhur Behl

Thesis: *Scenario2Vector: scenario description language based embeddings for traffic situations.*

Rodman Scholar (Top 5% of Engineering Class)

Research Experience

Data, Interpretability, Language and Learning Lab (DILL), USC

Los Angeles, CA

GRADUATE RESEARCH ASSISTANT

Advisor: Prof. Swabha Swayamdipta

My research focuses on human-centered AI alignment, with an emphasis on how AI systems can adapt to domain-specific tasks in high-impact societal applications. I study how domain experts and AI can be effective partners in information extraction and, more broadly, in applications for social good. To this end, I develop new methods and human-grounded evaluations designed to enable outcomes with real societal impact in social services, and healthcare by collaborating closely with community partners and stakeholders.

Vision, Language, and Learning Lab, UVA

Charlottesville, VA

MACHINE LEARNING RESEARCH ASSISTANT

Advisor: Prof. Vicente Ordóñez

Analyzed how pretraining data, model architecture, and training setup affect gender biases in visual recognition systems. In collaboration with Columbia University (Prof. Baishakhi Ray). Thesis

Link Lab, UVA

Charlottesville, VA

MACHINE LEARNING RESEARCH ASSISTANT

Advisor: Prof. Madhur Behl

Built and analyzed the Traffic Scenario Similarity (TSS) dataset containing human similarity judgments over driving scenarios. Experimented with multimodal transformer models in PyTorch to label traffic videos and understand scenario similarity. [Project Site](#)

Aerospace Engineering Research Group, UVA

Charlottesville, VA

MACHINE LEARNING RESEARCH ASSISTANT

Advisor: Prof. David Sheffler

Developed a 3D-printed autonomous UAV platform (Raspberry Pi + Pixhawk) for fully autonomous mission execution. Prototyped machine learning pipelines in OpenCV for onboard object recognition and mission-specific visual communication.

Publications & Preprints

Multi-Turn Interactions Reveal Hidden Safety Failures in Mental Health Language Model Use

Jaspreet Ranjit, Brihi Joshi, Angel Hwang, Swabha Swayamdipta

ShowCAIS'26

[Paper Coming Soon!](#) · 🏆 Best Poster Award ShowCAIS'26

LLM-in-the-Loop Detection of Misinformation on Social Media: A Case Study in the Illegal Wildlife Trade of Pangolins

Hannah Murray, Jaspreet Ranjit, Trent Kobielusz, Jeena Mahajan, Swaminathan Chellappa, Augusto Rivas Costante, Bistra Dilkina

Under Review at MisD'26

[Paper Coming Soon!](#)

Are We Automating the Joy Out of Work? Designing AI to Augment Work, Not Meaning

Jaspreet Ranjit, Ke Zhou, Swabha Swayamdipta, Daniele Quercia

ACM Conference on Human Factors in Computing Systems (CHI) 2026

[Paper](#)

Uncovering Intervention Opportunities for Suicide Prevention with Language Model Assistants

Jaspreet Ranjit, Hyundong J. Cho, Claire J. Smerdon, Yoonsoo Nam, Myles Phung, Jonathan May, John R. Blosnich, Swabha Swayamdipta

Association for Computational Linguistics (ACL 2026), EAAMO'25, GenAI4Health Workshop at NeurIPS'25

[Paper](#) · 🏆 Runner Up for Best Oral Presentation at ShowCAIS'25

Will AI Agents Free Us From Meaningless Work? A Human-Centered Analysis

Davide Ghia, Jaspreet Ranjit, Tania Cerquitelli, Daniele Quercia

CHIWORK'26

[Paper Coming Soon!](#)

The Flourishing Paradox: When Knowledge Work That Supports Human Flourishing Is Highly Exposed to AI Agents

Davide Ghia, Jaspreet Ranjit, Tania Cerquitelli, Daniele Quercia

Under Review at AIES'26

[Paper Coming Soon!](#)

Side-by-side Comparison Amplifies Dialect Bias in Language Models

Kritee Kondapally, Claire Smerdon, Pooja Patel, Ogheneyoma Akoni, Jevon Torres, Jaspreet Ranjit, Matthew Finlayson, Swabha Swayamdipta

In proceedings at FAccT'26, EAAMO'25

[Paper](#) · 🏆 Best Responsible AI Impact Award at SoCal AI Responsibility Summit 2025

The heaviest data: Can AI tools alleviate the weight of suicide mortality research?

John Blosnich, Jaspreet Ranjit, Swabha Swayamdipta

American Journal of Public Health (AJPH), 2026

[Paper Coming Soon!](#)

OATH-Frames: Characterizing Online Attitudes Towards Homelessness via LLM Assistants

Jaspreet Ranjit, Brihi Joshi, Rebecca Dorn, Laura Petry, Olga Koumoundouros, Jayne Bottarini, Peichen Liu, Eric Rice, Swabha Swayamdipta

Empirical Methods in Natural Language Processing (EMNLP), 2024.

[Paper](#) · 🏆 Outstanding Paper Award at EMNLP'24, Best Poster at ShowCAIS 2024

Community-Engaged AI for Youth Homeless Services: A Perspective on Data, Equity and Opportunity

Eric Rice, Peichen Liu, Swabha Swayamdipta, Lindsay Young, Jaspreet Ranjit, Christian Quijano, Olga Koumoundouros, Laura Onash-Vera, Bill Tang, Brian Rice

Under Review at Journal of Technology in Human Services (2025)

[Paper Coming Soon!](#) ·

Variation of Gender Biases in Visual Recognition Models Before and After Finetuning

Jaspreet Ranjit, Tianlu Wang, Baishakhi Ray, Vicente Ordóñez

NeurIPS Workshop on Algorithmic Fairness through the Lens of Time, 2023.

[Paper](#)

Scenario2Vector: Scenario Description Language Based Embeddings for Traffic Situations

Aron Harder, Jaspreet Ranjit, Madhur Behl

ACM/IEEE 12th International Conference on Cyber-Physical Systems (ICCPS), 2021.

[Paper](#)

Awards & Press

2026

Best Poster Award at ShowCAIS 2026: Multi-Turn Interactions Reveal Hidden Safety Failures in Mental Health Language Model Use.

2025

Voted Judge's Favorite, and received Honorable Mention in AI for Social Good Hackathon hosted by LA Tech Week x Lovable

WiSE Merit Award for 2025

First Place in Min Family Challenge: CaseFlo – an agentic AI platform that streamlines the workflow of case managers, enabling effective discovery of resources, reduced case worker burnout, and data-driven interventions for people experiencing homelessness.

Runner up for best oral presentation at ShowCAIS 2025: Designing and Validating Intervention Opportunities for Suicide Prevention with Language Model Assistants.

2024

Viterbi Magazine: Featured in the USC Viterbi Magazine under the theme: “Human Centered” The USC School of Advanced Computing.

Received an outstanding paper award at EMNLP 2024 for OATH-Frames

Passed my Qualifying Exam

Presidential Leadership Council: Invited to present at USC's Presidential Leadership Council.

Spectrum News Coverage: Spectrum News covers our work on OATH-Frames: Characterizing Online Attitudes Towards Homelessness via LLM Assistants.

USC Media Coverage: USC covers our work on OATH-Frames: Characterizing Online Attitudes Towards Homelessness via LLM Assistants.

Best Poster Award: ShowCAIS 2024: Awarded best poster at annual conference organized by Center for AI in Society.

2021

Rodman Scholar: Academic honor awarded to top 5% of undergraduate engineering class at University of Virginia.

Cornell, Maryland, and Max Planck Pre-Doctoral Research School: Promising undergraduate and Masters students are invited to attend this program that provides an overview of the state of the art research in Computer Science.

Truly OpenML: Led a team of four people to pitch a web application that provides a collaborative, intuitive and accessible platform for individuals who are passionate about learning machine learning (ML). Semi-finalist at the American Evolution Innovator's Cup.

Invited Talks

2025

National Alliance to End Homelessness x Center for Homelessness Impact: Using AI to End Homelessness

Guest Lecture in AI for Sustainable Development CSCI-461 at USC on 'Human-AI Collaboration: LM Applications for Social Good'

Homelessness Policy Research Institute Research Seminar

Google: Technology, AI, Society and Culture Team: STAR Talk Series

2024

National Alliance to End Homelessness: Research Seminar

Service & Leadership

Conference Reviewer

ShowCAIS Symposium– 2026
International Joint Conference on Artificial Intelligence (IJCAI)– 2026
ACM Conference on Fairness, Accountability, and Transparency (FAccT)– 2026
ACM Conference on Human Factors in Computing Systems (CHI)– 2026
Workshop on Natural Language Processing for Positive Impact (NLP4PI)– 2025
International Joint Conference on Artificial Intelligence (IJCAI)– 2025
ShowCAIS Symposium– 2025
Empirical Methods in Natural Language Processing (EMNLP)– 2024
Association for Computational Linguistics (ACL)– 2024

Leadership

Secretary, Women in Science and Engineering (WiSE)– USC
Student Leader, Center for AI in Society (CAIS)– USC

Organizing

ShowCAIS 2026 Symposium– USC
ShowCAIS 2025 Symposium– USC

Teaching

Fairness in AI 2026 (Andrew Gordon)– USC
AI for Sustainable Development 2025 (Bistra Dilkina)– USC

Mentoring

Graduate Mentoring

Augusto Rivas Costante, USC

Uncovering Intervention Opportunities for Suicide Prevention with Language Model Assistants: 2025-present

Swaminathan Chellappa, USC

Uncovering Intervention Opportunities for Suicide Prevention with Language Model Assistants: 2025-present

Risha Surana, USC →Microsoft

Aspect Based Summarization for Public Attitudes Towards Homelessness: 2024-present

Claire J. Smerdon, USC →Palo Alto Networks

Side-by-side Comparison Amplifies Dialect Bias in Language Models: 2024-2026

Kritee Kondapally, USC

Side-by-side Comparison Amplifies Dialect Bias in Language Models: 2024-2026

Pooja Patel, USC

Side-by-side Comparison Amplifies Dialect Bias in Language Models: 2024-2026

Ogheneyoma Akoni, USC

Side-by-side Comparison Amplifies Dialect Bias in Language Models: 2024-2026

Myles Phung, USC →Citadel Securities

Uncovering Intervention Opportunities for Suicide Prevention with Language Model Assistants: 2024-2025

Ruyuan Zhou, USC →PhD at Georgia Tech

OATH-Frames: Characterizing Online Attitudes Towards Homelessness via LLM Assistants: 2023-2024

Undergraduate Mentoring

Aryan Gulati, USC →Bloomberg

Aspect Based Summarization for Public Attitudes Towards Homelessness: 2024-2025

Catherine He, USC

Aspect Based Summarization for Public Attitudes Towards Homelessness: 2024-2025

Industry Experience

Johns Hopkins / Carnegie Mellon: Data Science for Social Good

RESEARCH FELLOW

Mentor: Rayid Ghani

Coming soon!

Baltimore, MD

May 2026 – Aug 2026

Nokia Bell Labs: Social and Responsible AI Team

RESEARCH INTERN

Mentor: Daniele Quercia, Ke Zhou

Conducted mixed-methods research on human-AI alignment in workplace automation, combining survey design, behavioral analysis, and language-model simulations. Modeled divergences between worker and developer automation preferences to inform human-centered frameworks for responsible and value-aligned AI deployment. Investigated how AI-driven workplace automation reshapes workers' sense of value, purpose, and meaning. Designed and conducted large-scale surveys with developers and workers to examine divergences in automation preferences, and used language-model simulations to scale behavioral insights. Developed computational frameworks for socially informed automation that embed human well-being, stakeholder values, and cultural context into AI system design.

Cambridge, UK

Jun 2025 – Aug 2025

Vimeo

MACHINE LEARNING RESEARCHER ON SEARCH AND RECOMMENDATIONS

Mentor: Silvena Chan

Analyzed gender biases in search and recommendation models and formulated a bias identification framework with the Rank Bias metric quantifying gender biases in ranked search results. Developed learning to rank (LTR) models in Pytorch using RankNet and LambdaMART, and developed an internal dataset for LTR models in private search. Worked with big data in Snowflake and wrote queries in SQL to scale bias experiments.

Medium Publication

New York, NY

Jun 2021 – Aug 2021

Expedition Technology

MACHINE LEARNING ENGINEER

Mentor: Cheryl Daner

Researched anchorless object detection techniques for 3D point cloud object detection. Designed a convolutional neural network on the basis of existing VoxelNet and CenterNet architectures in Tensorflow. *Summary of Work*

Herndon, VA

Jun 2019 – Aug 2019

NASA Goddard Spaceflight Center

CORE FLIGHT SOFTWARE ENGINEER

Mentor: Alessandro Geist

Developed and benchmarked core Flight Software apps in C/C++ that directed AI image processing and command/telemetry with ground station. Worked with Xilinx Platform Studio and ISE Design Suite.

Greenbelt, MD

Jun 2018 – Aug 2018

NASA Langley Research Center

3D PRINTING ENGINEER

Mentor: Godfrey Sauti

Leveraged sensor technology to design and improve the dimensional integrity of a printed component using Pronterface. Designed 3D components in Inventor.

Hampton, VA

Jun 2016 – Aug 2016